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Educational Paper

Organization of nutritional care

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SUMMARY

The increasingly complex health care systems, the large and diverse population to whom nutritional therapy is addressed, and the need of specialized health care professionals implies an organizational strategy to be defined. The organization of nutritional care is a fundamental strategy to promote an optimal, quality and safety nutritional therapy. This is why, the organization of nutritional care should be based on 4 principles, including: Nutritional care should be considered as a process, a multidisciplinary approach is essential to a good nutritional care, the economic approach and the patient centered approach are essential to the organization of nutritional care. In this paper we will address the organization of nutritional care and its 4 principles.

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Learning objectives

1. To identify the general principles of nutritional care
2. To understand the importance of the nutritional care process
3. To recognize the benefits of a multidisciplinary approach to nutritional care

Key messages

- Nutrition care processes such as screening, prescribing nutrition treatment, monitoring daily nutrition intake and recording of malnutrition status are applied in less than 50 % of patients.
- The organization of nutritional care is a fundamental strategy to promote an optimal, quality and safety nutritional therapy.
- The delivery of nutritional care should be based in 4 principles:
  1) Nutritional care should be considered as a process;
  2) A multidisciplinary approach is essential to a good nutritional care;
  3) The economic approach should be integrated to the organization of nutritional care;
  4) The patient centered approach is essential to the organization of nutritional care.
- Patient-centred care is about treating a person receiving healthcare with dignity and respect.

1. Introduction

Nutrition therapy (i.e. diets, oral nutrition supplements, enteral and parenteral nutrition) is a cost-effective therapy that has been associated with better outcomes including improved survival and non-elective hospital readmission rates among medical inpatients who were malnourished [1–3]. However, not all patients benefit form nutrition therapy as it has been shown by the nutritionDay data. According to data from 2009 to 2015 of 155,000 patients worldwide, only 29 % of patients identified at nutritional risk received any kind of nutrition therapy [4]. Nearly 50 % of in-patients with a history of weight loss did not have their nutrition intake monitored, a nutrition expert was never consulted in more than 46 % of patients even with 3 risk factors [5]. Thus, the lack of delivery and access to nutritional care is a worldwide problem. Different reasons and strategies have been evocated to address this

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Abbreviations

BMI	Body Mass Index
CLAN	Comité de Liaison Alimentation Nutrition
CNSU	Clinical Nutrition Support Unit
CNU	Clinical nutrition units
DRM	Disease-related malnutrition
DST	Disease-specific Support Teams
ESPEN	European Society for Clinical Nutrition and Metabolism
FFM	Fat free mass
FM	Fat mass
FREDA	fairness, respect, equality, dignity, and autonomy
GLIM	Global Leadership Initiative on Malnutrition
HGS	Hand grip strength
HPN	Home Parenteral Nutrition
MNA	Mini Nutritional Assessment
MNA-SF	Mini Nutritional Assessment short form
MS	Medical Schools
MST	Malnutrition Screening Tool
MUST	Malnutrition Universal Screening Tool
NSC	Nutrition Steering Committee
NST	Nutrition Support Team
QoL	Quality of life
SNAQ	Short Nutritional Assessment Questionnaire
SPPB	Short Physical Performance Battery
WHO	World Health Organization

issue and the consequent high prevalence of disease-related malnutrition (DRM) [5].

The organization of nutritional care is a fundamental strategy to promote an optimal, quality and safety nutritional therapy. The increasingly complex health care systems, the large and diverse population to whom nutritional therapy is addressed, and the need of specialized health care professionals implies an organizational strategy to be defined.

The organization of nutritional care should be based on 4 principles:

1) Nutritional care should be considered as a process.
2) A multidisciplinary approach is essential to a good nutritional care.
3) The economic approach should be integrated to the organization of nutritional care.
4) The patient centered approach is essential to the organization of nutritional care

In this paper, we will address the organization of nutritional care and its 4 principles.

2. Nutritional care is a process

According to The European Society for Clinical Nutrition and Metabolism (ESPEN) guidelines, nutritional care is a process that "should be provided in a systematic sequence that involves distinct interrelated steps".

These steps include malnutrition risk screening, diagnosis and assessment of malnutrition, nutrition therapy and monitoring. The implementation of nutritional care should cover the primary care level, the inpatients and out patients. (Fig. 1).

2.1. Malnutrition risk screening

Malnutrition risk screening is defined by ESPEN as "a rapid process performed to identify subjects at nutritional risk, and should be performed using an appropriate validated tool in all subjects that come in contact with healthcare services." It is recommended that screening should be performed within the first 24–48 h after first contact and thereafter at regular intervals [6]. The screening of at-risk patients will serve to identify those patients that will benefit from nutritional therapy in terms of better outcomes (i.e. morbidity, mortality, economic burden, quality of life). Schuetz et al. within the EFFORT trial [3], proved that feeding patients at nutritional risk (large, heterogenous population of medical patients) with an individualized nutritional algorithm is beneficial with lowering risks for adverse outcome, mortality and functional decline. There are several risk-screening tools in use. All tools use similar criteria for identification (weight loss, BMI, etc), whereas methodologies, selection of criteria and cut-offs vary.

ESPEN recommend the use of Nutrition Risk Screening-2002 (NRS- 2002) and the Malnutrition Universal Screening Tool (MUST) and for the elderly the Mini Nutritional Assessment (MNA) either in its full or short form (MNA-SF) [7,8]. Other validated tools, which are frequently used, include the Malnutrition Screening Tool (MST) and the Short Nutritional Assessment Questionnaire (SNAQ). For sarcopenia screening, the ESPEN endorsed statement from the European Working Group on Sarcopenia in Older Persons recommends screening for sarcopenia from age 65 years onwards by

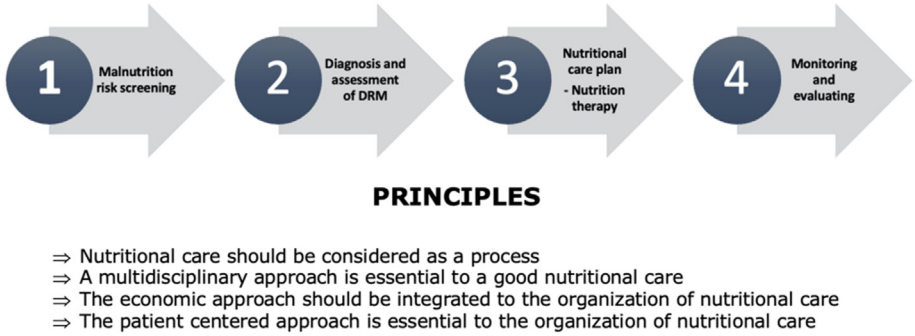

Fig. 1. Nutritional care is a process that should be provided in a systematic sequence that involves 4 distinct interrelated steps, and should be based in 4 principles. DRM: disease-related malnutrition.

measuring handgrip strength and/or chair stand test, and for assessment to measure muscle mass (quantity and quality) [9].

Subjects identified at nutritional risk need to undergo nutritional assessment.

2.2. Diagnosis and assessment of disease-related malnutrition

Diagnosis and assessment should be performed in all subjects identified as being at risk by nutritional risk screening. The diagnosis of DRM is the basis for determining a nutritional treatment plan. ESPEN Recommend the Global Leadership Initiative on Malnutrition (GLIM) criteria for diagnosis of malnutrition.

The GLIM consortium composed of expert from the major global clinical nutrition societies (i.e. ASPEN (USA), ESPEN (Europe), FELANPE (Latin America) and PENSA (Asia)) started in 2016 and defined the criteria and a framework to diagnose malnutrition. These includes phenotypic criteria (i.e., weight loss, low BMI, and reduced muscle mass) and etiologic criteria (i.e., reduced food intake or assimilation and high disease burden/inflammation). GLIM propose a 2-step procedure [10–12].

- Step 1: is screening by use of any well-validated screening tool. Individuals with risk of malnutrition should go to step 2.
- Step 2: be assessed by the five diagnostic criteria.

The diagnosis of disease-related malnutrition is confirmed when at least one of the phenotypic criteria combined with at least one of the etiologic criteria is aberrant from the suggested cut-offs. The severity (i.e., moderate or severe) is assessed based on the degree of deviation from normal (considering the phenotypic criteria).

GLIM criteria seem to have a strong prognostic value of regarding adverse clinical outcomes and other long-term outcomes. However, the ability of GLIM criteria to predict therapeutic response to nutritional interventions needs further research [13].

Based on the underlying etiology malnutrition is categorized as related to [6]:

- DRM-chronic disease with inflammation
- DRM-chronic disease with minimal or no perceived inflammation
- DRM-acute disease or injury with severe inflammation
- Malnutrition related to starvation/food shortage due to socio-economic or environmental factors.

Currently, the WHO ICD-11 terminology for malnutrition is based only on the body mass index (BMI <18kg/m2), but should evolve to include the current conception of DRM [14].

2.3. Nutritional care plan

According to ESPEN the nutritional care plan is “a scheme for nutrition therapy based on the results of the assessment. A comprehensive nutritional care plan defines the rationale, explains the nutrition therapy and provides suggestions for monitoring the efficacy of the plan and reassessment” [6].

The nutritional care plan includes information on:

- Energy, nutrient and fluid requirements
- Measurable nutrition goals (immediate and long-term)
- Instructions for implementing the specified form of nutrition therapy
- The most appropriate route of administration and method of nutrition access
- Anticipated duration of therapy
- Monitoring and assessment parameters
- Discharge planning and training at home (if appropriate)

This plan should be developed by a “multi/interdisciplinary team together with the patient and the caregiver in order to achieve patient centered treatment goals”.

2.3.1. Nutrition therapy

Nutrition therapy describes “how nutrients are provided to treat any nutritional-related condition. Nutrition or nutrients can be provided orally (regular diet, therapeutic diet, e.g. fortified food, oral nutritional supplements), via enteral tube-feeding or as parenteral nutrition to prevent or treat malnutrition in an individualized way” [6]. Evidence shows that nutritional therapy should be ideally personalized [15]. The problem of DRM is far more complex than a simple issue of lack of adequate nutrient intake. Indeed, according to Cardenas and Ochoa “the problem may lie in the fact that different illnesses (and different patients) more than likely exhibit unique metabolic alterations that demand personalized interventions”. Such is the case in cancer patients or critical illness where large clinical trials recruiting heterogenous populations have studied the possible benefit of single nutrient supplementation as if all patients exhibited the same metabolic requirements [16].

The concept of personalized nutrition is based on data that emerge from studies showing that some populations of patients can benefit from nutritional therapy interventions and others that need a different approach [17]. Several secondary analyses of the EFFORT trial suggested that patients most strongly responding to individualized nutritional interventions include:

- Patients with impaired kidney function: admission estimated glomerular filtration rates (eGFR) was a strong predictor for the beneficial effects of nutritional support in regard to lowering of 30-day mortality [15].
- Patients with low hand grip strength (HGS): each decrease of 10 kg in HGS was associated with increased risk of 30-d mortality (female: adjusted OR: 2.11; 95 % CI: 1.23, 3.62, P = 0.007; male: adjusted OR: 1.44; 95 % CI: 1.07, 1.93, P = 0.015) and 180-d mortality (female: adjusted OR: 1.45; 95 % CI: 1.0, 2.10, P = 0.048; male: adjusted OR: 1.55; 95 % CI: 1.28, 1.89, P < 0.001). In patients with low HGS, individualized nutritional support was most effective in reducing mortality (adjusted OR: 0.29; 95 % CI: 0.10, 0.82 in patients in the ≤10th percentile compared with OR: 0.98; 95 % CI: 0.66, 1.48 in patients in the >10th percentile; P for interaction = 0.026) [18].
- Patients in hibernating state (with low T3 syndrome). EFFORT trial secondary analysis suggests that medical inpatients at nutritional risk with low T3 syndrome have an increase in mortality and may show a more pronounced beneficial response to nutritional support interventions. Nutritional support only lowered mortality in the group of patients with low T3 syndrome (adjusted odds ratio of nutritional support of 0.82 [95%CI 0.47 to 1.41] vs. 1.47 [95%CI 0.55 to 3.94]) [19].

EFFORT secondary analyses also suggest that patients not responding to individualized nutritional interventions include:

- Patients with a high inflammatory state C-reactive protein (CRP) > 100 mg/dl) [20,21].
- Albumin and prealbumin levels seem not being helpful for guiding nutritional interventions [22,23],
- Higher or lower severity of malnutrition risk according to malnutrition screening tools [24],
- Low muscle mass as assessed by CT scans [25].

2.3.2. Hospital food intake

ESPEN recommends actions regarding the organization of food catering, the prescriptions and indications of diets, as well as monitoring of food intake at hospital, rehabilitation center, and nursing home, all of these by taking into account the patient perspectives; and propose a systematic approach to adapt the hospital food to the nutritional status and potential food allergy or intolerances [26]. The aim is to improve coverage of nutritional needs by hospital food, and reduce the risk of malnutrition and its related complications.

2.4. Monitoring and evaluating the effects of nutritional care and therapy documentation

According to ESPEN, monitoring of nutrition therapy is "a measure to check and adjust that nutrition delivery is in progress and nutrition intake or provision is sufficient, as well as to assure tolerance and that goals and expected outcomes are achieved. The monitoring procedures require an individual plan where nutrition goals are defined" [6].

- Nutrition provision and intake: Are calculated requirements of fluid, energy and protein met?
- Weight, anthropometry, body composition: Does e.g. weight, fat free mass (FFM) or fat mass (FM) change as expected?
- Biochemistry: There are no good biochemical markers of the nutritional status. Plasma albumin and transthyretin/prealbumin concentrations may be used mainly to indicate and monitor catabolic activity. Their validity as nutrition indicators is low in view of their perturbation by inflammation.
- Function: e.g. hand grip strength (HGS), chair rise tests and gait speed, either alone or combined in the Short Physical Performance Battery (SPPB) [35] could be used.
- Quality of life (QoL): e.g. EQ-5D or HRQOL, or other tools relevant to the diagnosis, could be used.

3. The multidisciplinary approach

The multidisciplinary approach is essential to a good nutritional care. This implies a diverse group of professionals working together. In clinical nutrition traditionally dietitians, doctors, nurses and pharmacists have collaborated to delivery nutritional care. New professionals are now joining nutritional care as for example physiotherapists, speech therapists, and psychologists.

The benefits of adopting a multidisciplinary approach include for patients: "a) shorter timeframes from diagnosis to treatment, b) greater likelihood of receiving care in accord with clinical practice guidelines, including psychosocial support, c) increased access to information d) improved satisfaction with treatment and care.

For health professionals: a) improved patient care and outcomes through the development of an agreed treatment plan, b) streamlined treatment pathways and reduction in duplication of services, c) improved coordination of care, d) educational opportunities for health professionals, e) improved mental well-being of health professionals" [27].

According to ESPEN, providing nutritional care can be organized in different ways [6]:

a) Care catering/Hospital catering
b) Nutrition Steering Committee
c) Disease-specific Support Teams
d) Clinical Nutrition Care Unit
e) Clinical Nutrition Support Unit
f) Nutrition Support Team

3.1. Care catering/hospital catering

According to ESPEN a care catering or hospital catering is "the provision of menu services (in-house or outsourced) in health care facilities. The minimum requirements of hospital and care catering are to serve a variety of foods that are suitable and adapted to all types of patients with a variety of energy and nutrient densities" (6).

ESPEN recommend that each hospital shall have a structured hospital food facility consisting of a kitchen, a delivery system, and an ordering system [26].

Cultural, educational, religious and food taste preferences should be considered and at least should be taken into account when proposing the menu choice to the patient. Patients at risk or malnourished should have special attention by considering individual choices with respect to food items and portion sizes have to be ensured.

3.2. Nutrition steering committee (NSC)

According to ESPEN a NSC is a "committee at a hospital or care facility of a mixed interdisciplinary composition including directors, managers, health professionals and catering staff" [6]. The main objective of a NSC is "to set standards for the structure, procedures and management of clinical nutrition at the relevant institutions. Depending on the legal status of the NSC (mainly decided by the care facility management) it may also be responsible for the audit of the nutritional care and responding to nutritional incidents."

As a matter of example, in France, it is strongly recommended for each hospital to have a NSC called « CLAN « (Comité de Liaison Alimentation Nutrition). The CLAN's mission is to improve nutritional care of admitted patients. To this end, the CLAN implements actions to help solve all problems concerning diet or nutritional care of patients within the hospital. This involves monitoring the quality of food and catering in hospitals by ensuring that the safety criteria are followed, using the HACCP method (Hazard Analysis Critical Control Point). Promoting the screening and management of malnutrition in hospitalized patients is also a major role of the CLAN. The composition of the CLAN varies according to the hospital size. It includes at least a medical doctor, a nurse, a dietician, a catering manager. A pharmacist, a medical training manager, a quality manager and a hygiene manager can also participate in the CLAN work.

3.3. Disease-specific support teams (DST)

DST are teams focused to specific care facilities as for example, Obesity. A Obesity team is a multidisciplinary team of specialists consisting of "physicians, dietitians, nurses, physio-therapists, behavioral therapist (psychologist/psychiatrist) as well as other relevant professionals" (6). The Obesity team provides personalized, patient-centred and comprehensive weight man-agement/lifestyle programmes which take into account the comorbidities of the obese patients. The Obesity team should also assist "bariatric surgery" services for pre- and post-operative care. Other diseases and conditions can also benefit from DST as diabetes, chronic obstructive pulmonary disease, cancer and palliative care.

In France, the CSO "Centres Spécialisé d'Obésité" (specialized obesity center) were created in 2012 by the french government to fulfill two main missions. Their first mission is the multidisciplinary management of severe and/or complex obesity, through multiple expertise (nutrition, endocrinology, psychology, surgery). Multidisciplinary consultation meetings are organized for obese patients referred to bariatric surgery. Their second mission is the

organization, animation and development of medical care for obese patients in the different regions of France. There are 37 CSO in France.

3.4. Clinical nutrition care unit and clinical nutrition support unit (CNSU)

Clinical nutrition care (delivery by a dietitian) or CNSU are specific wards managed by a multidisciplinary team of specialized physicians, nurses, dietitians and pharmacists [6]. This type of organization can vary according to each hospital's organization. In the CNSU patients that require nutrition therapy or receive home artificial/medical nutrition who develop metabolic complications, or other type of complications such as central line infection can be hospitalized in specific clinical nutrition support wards managed by a multidisciplinary team of specialized professionals.

The first clinical nutrition units (CNUs) emerged in the public hospitals of the Spanish National Health System (NHS) in the 80s and have gradually been incorporated into the Departments of Endocrinology and Nutrition. In a study published in 2017 on a random sample of 20 % of Spanish hospitals of the National Health System showed that well defined CNUs existed in 51 % of the centres, and other NSTs in 14.3 % of the hospitals, with less independence. In a minority of hospitals, the nutritional treatment directly depends on the physician responsable of the patient. CNUs are more common in hospitals with more than 400 beds. Only 50 % of CNUs or NSTs had a physician working with them full time, and just 40 % had a pharmacist. This percentage was even lower for all other professional categories, except for nurses. What was also clear was that the existence of well-organized structures can be accompanied by benefits with a direct impact on care quality [28].

3.5. Nutrition support team (NST)

A NST is a multi-disciplinary team of physicians, dietitians, nurses and pharmacists. Other relevant professionals may also be part of the NST, e.g. physiotherapist and speech therapists. In contrast to the CNSU, the NST is not a specific ward but usually works across all hospital wards, disease or condition to delivery nutrition care to pediatric and adult patients.

The main objective of the NST is to “support hospital staff in the provision of nutrition therapy, especially enteral or parenteral nutrition, to ensure that the nutritional needs of patients are satisfied, especially for those patients with complicated nutritional problems. Moreover, the objective includes ensuring that all nutrition therapy utilises state-of-the-art knowledge and techniques to prevent and treat disease-related malnutrition of both inpatients and out-patients” (6). According to A Barrocas, NSTs were established in the early 1970's in response to the development of parenteral nutrition (PN) by Dudrick and colleagues.

3.6. Home artificial nutrition

The organization of nutritional care at home setting differs from hospitalized patients because there is a shift in primary responsibility from health care professionals to patients and caregivers [28]. According to the ESPEN Guidelines on Home Parenteral Nutrition (HPN) the patients on HPN should “be cared for by specialized, dedicated and a clearly identifiable hospital unit, normally termed HPN center or Intestinal Failure center or intestinal rehabilitation center”. According to the “suitability of the home care environment should be assessed and approved by the HPN nursing team before starting HPN, wherever possible.” [29].

All healthcare professionals who are directly involved in patient care should receive education and training, relevant to their duties, on the different aspects related to the safe provision of HEN and the importance of providing adequate nutrition. In hospitals who discharge patients with HEN should employ at least one specialized nutrition support nurse or dietician. Ideally, these hospitals should have a NST working within the clinical governance framework.

According to ESPEN, hospitals who discharge patients with home enteral nutrition should have NST working within the clinical governance framework and at least one specialized nutrition support nurse or dietician [30]. Healthcare professionals should ensure that all people who need nutrition support receive coordinated care from a multidisciplinary NST [30].

4. Clinical nutrition training of doctors and other health care professionals

Nutrition education in Medical Schools (MS) has been neglected for many years for multiples reasons including the lack of time in the medical curriculum, resulting in lack of awareness of the importance of nutrition-related disorders of the future doctors. An ESPEN survey in 2017 showed that in most of the MS the number of hours devoted to nutrition was inadequate, and 83.4 % agreed to include at least 8 h in the curriculum [31]. Based on this survey, ESPEN and several European Medical Schools agreed on a minimum curriculum content that should be mandatory to all MS, covering the 3 domains of human nutrition (basic nutrition-applied nutrition-clinical nutrition), with its learning objectives and more than 20 different topics identified [32]. These contents can be taught in different models, according to the MS, time in the curriculum, and faculty in each center. We identified 4 models: vertical integration, horizontal integration, mandatory nutrition subject, and elective nutrition subject, that can be combined in some MS, according to their needs [31–33].

5. The economic approach should be integrated to the organization of nutritional care

Disease-related malnutrition leads to an increase in hospital costs [35]. In-hospital nutritional therapy is a highly cost-effective intervention [2] to reduce risks for ICU admissions and hospital-associated complications, while improving patient survival [36–39]. An additional individual cost of 1640 € to 5829 € has been identified for European hospitalized patients. Moreover, at the national level, the costs of disease related malnutrition ranged between 32.8 million € and 1.2 billion €., this means 2.1 % and 10 %.

6. The patient-centred care approach

Independent of any type of nutritional care organization, the patient centred approach would be at the core of the patient's care. Also known as ‘person-centred care’, this approach is essential to the organization of nutritional care. Patient-centred care is about respecting the individual preferences and diversity. It is also about involving patients in all decisions about their health, and the needs and the respects of rights right to make health decisions and choices. Patient-centred care is about treating a person receiving healthcare with dignity and respect.

Ethics and human rights are guiding values for clinical nutrition practitioners [34]. Together they ensure a patient-centered approach, where the needs and rights of the patients are of the most significant importance. Human rights are based on the powerful idea of equal dignity for all people while expressing a set of core values, including fairness, respect, equality, dignity, and autonomy (FREDA). Through the analysis of FREDA principles, we have provided the elements to understand human rights and how a HRBA can support clinicians in the decision-making process. In the

modules 11.2 and 11.3 Ethics, human rights and legal aspects in clinical nutrition will be addressed.

7. Conclusion

The organization of nutritional care is a fundamental strategy to promote an optimal, quality and safety nutritional therapy. The increasingly complex health care systems, the large and diverse population to whom nutritional therapy is addressed, and the need of specialized health care professionals implies an organizational strategy to be defined. This is why, the organization of nutritional care should be based on 4 principles, including: Nutritional care should be considered as a process, a multidisciplinary approach is essential to a good nutritional care, the economic approach and the patient centered approach are essential to the organization of nutritional care.

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The authors declare that they have no competing financial interest or personal relationships that could have appeared to influence the work reported in this paper.

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